

2) K-Maps for 2 logical expressions f_1 & f_2 are given.

BC	00	01	11	10
A				
0		1		
1		1		1

f_1

BC	00	01	11	10
A				
0	1	1		
1	1			1

f_2

Perform (a) $f_1 f_2$ (c) $f_1 \oplus f_2$
(b) $f_1 + f_2$ (d) $f_1 \odot f_2$

Ans (a) $f_1 = \sum m(1, 5, 6)$

$f_2 = \sum m(0, 1, 4, 6)$

$f_1 \cdot f_2 = \sum m(1, 6)$

(b) $f_1 + f_2 = \sum m(0, 1, 4, 5, 6)$

(c) $f_1 \oplus f_2 = \sum m(0, 4, 5)$

(d) $f_1 \odot f_2 = \sum m(1, 2, 3, 7, 6)$

$$A \begin{array}{c|cccc} BC & 00 & 01 & 11 & 10 \\ \hline 0 & & 1 & & \\ 1 & & & & 1 \end{array}$$

$$A \begin{array}{c|cccc} BC & 00 & 01 & 11 & 10 \\ \hline 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{array} \quad A \begin{array}{c|cccc} BC & 00 & 01 & 11 & 10 \\ \hline 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array}$$

$\Rightarrow f_1 \cdot f_2$

(*** cell to cell operation is performed) ..

$\Rightarrow$ 5-Variable $\times$ K-Map :-

5-Variables $\Rightarrow 2^5 = 32$ cells.

$f(A, B, C, D, E) \rightarrow$

$$A \begin{array}{c|cccc} DE & 00 & 01 & 11 & 10 \\ \hline BC & 00 & 01 & 11 & 10 \\ 00 & 0 & 1 & 3 & 2 \\ 01 & 4 & 5 & 7 & 6 \\ 11 & 12 & 13 & 15 & 14 \\ 10 & 8 & 9 & 11 & 10 \end{array} \quad A \begin{array}{c|cccc} DE & 00 & 01 & 11 & 10 \\ \hline BC & 00 & 01 & 11 & 10 \\ 00 & 16 & 17 & 19 & 18 \\ 01 & 20 & 21 & 23 & 22 \\ 11 & 28 & 29 & 31 & 30 \\ 10 & 24 & 25 & 27 & 26 \end{array}$$

$A=0$

$A=1$

$\Rightarrow$ In adjacent cells, only one bit will be different. $9 \rightarrow 01001$

$25 \rightarrow 11001$

$\Rightarrow$ Mapped Pairs are also adjacent to each other.

(9) $f(A, B, C, D, E) = \sum m(3, 4, 6, 11, 12, 14, 15, 19, 23, 25, 28, 30)$

$$A \begin{array}{c|cccc} DE & 00 & 01 & 11 & 10 \\ \hline BC & 00 & 01 & 11 & 10 \\ 00 & & & 1 & \\ 01 & 1 & & & 1 \\ 11 & 1 & & & 1 \\ 10 & & & 1 & \end{array} \quad A \begin{array}{c|cccc} DE & 00 & 01 & 11 & 10 \\ \hline BC & 00 & 01 & 11 & 10 \\ 00 & & 1 & & \\ 01 & 1 & & & 1 \\ 11 & 1 & & & 1 \\ 10 & & 1 & & \end{array}$$

$$Y = \bar{A}(C\bar{E} + \bar{C}DE)$$

$$Y = A(C\bar{E} + \bar{C}DE)$$

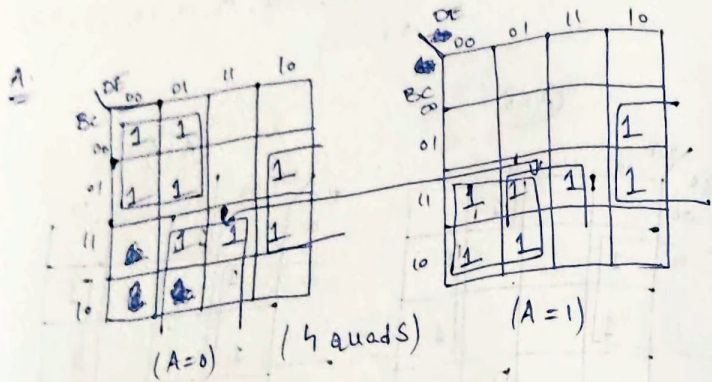
$$Y = \bar{A}\bar{C}\bar{E} + \bar{A}\bar{C}DE + A\bar{C}\bar{E} + A\bar{C}DE$$

$$Y = C\bar{E}(A + \bar{A}) + \bar{A}\bar{C}DE + A\bar{C}DE$$

$$\Rightarrow Y = C\bar{E} + \bar{A}\bar{C}DE + A\bar{C}DE$$

$$(Q) f(A, B, C, D, E) = \bar{A}\bar{B}\bar{D} + BCE + CDE + AB\bar{D}$$

Draw K-Map.



$$(E+D+\bar{E}) \cdot (\bar{B}+C+\bar{E}) \cdot (\bar{B}+C+\bar{D}) \cdot (E+\bar{D}+\bar{E}) \cdot (\bar{A}+B+C+\bar{D}) \cdot (B+C+\bar{D}+E)$$

(*) K-Maps of XOR Gate, XNOR Gate and their Complements :-

$$f(A, B) = \bar{A}B + A\bar{B} \rightarrow \text{XOR}$$

A \ B	0	1
0	0	1
1	1	0

$$f(A, B) = AB + \bar{A}\bar{B} \rightarrow \text{XNOR}$$

A \ B	0	1
0	1	0
1	0	1

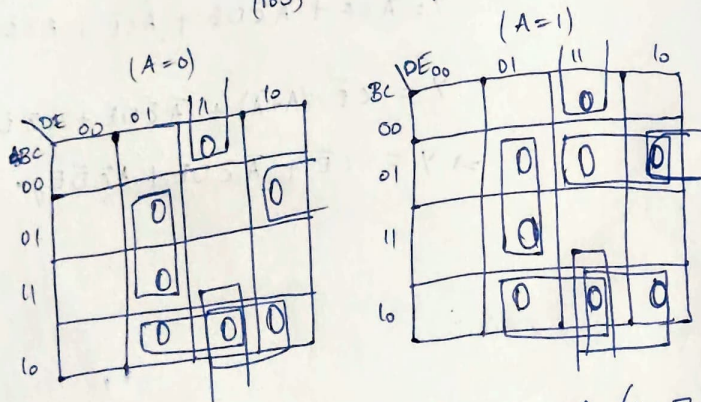
$$\Rightarrow \begin{array}{c|cccc} BC & 00 & 01 & 11 & 10 \\ \hline A=0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{array} \rightarrow A \oplus B \oplus C = A \odot B \odot C$$

$$\Rightarrow \begin{array}{c|cccc} BC & 00 & 01 & 11 & 10 \\ \hline A=0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{array} \rightarrow \begin{aligned} A \oplus B \oplus C &= A \odot B \odot C \\ &= A \odot B \odot C \\ &= B \odot A \odot C = A \odot B \odot C \end{aligned}$$

A.

$$(Q) f(A, B, C, D, E) = \pi M (3, 5, 6, 9, 10, 11, 13, 14, 21, 23, 22, 25, 26, 27, 29)$$

(Pos)



$$Y = (\bar{A} + \bar{C} + D + \bar{E}) \cdot (A + \bar{C} + D + \bar{E}) \cdot (\bar{A} + \bar{B} + C + \bar{E}) \cdot (A + \bar{B} + C + \bar{E})$$

$$(\bar{A} + \bar{B} + C + \bar{D}) \cdot (A + \bar{B} + C + \bar{D}) \cdot (\bar{A} + C + \bar{D} + \bar{E})$$

$$(A + C + \bar{D} + \bar{E}) \cdot (\bar{A} + B + \bar{C} + \bar{D})$$

⇒

	CD	00	01	11	10
AB	00	1	0	1	0
	01	1	0	1	0
	11	1	0	1	0
	10	1	0	1	0

$$\Rightarrow A \oplus B \oplus C \oplus D //$$

⇒

	CD	00	01	11	10
AB	00	1	0	1	0
	01	0	1	0	1
	11	1	0	1	0
	10	0	1	0	1

$$\Rightarrow \overline{A \oplus B \oplus C \oplus D} = A \oplus B \oplus C \oplus D$$

(a) The K-Map given below shows :-

$(A=0)$

	DE	00	01	11	10
BC	00	0	1	0	1
	01	1	0	1	0
	11	0	1	0	1
	10	1	0	1	0

$(A=1)$

	DE	00	01	11	10
BC	00	0	1	0	1
	01	1	0	1	0
	11	0	1	0	1
	10	1	0	1	0

★

(a) $A \oplus [B \oplus C \oplus D \oplus E]$ (b) $B \oplus C \oplus D \oplus E$ (c) Both (d) None

★

$$[B \oplus C \oplus D \oplus E] (A + \bar{A}) = \underline{B \oplus C \oplus D \oplus E} [C].$$

The K-Map given below shows :-

(a)

$(A=1)$

	DE	00	01	11	10
BC	00	1	0	1	0
	01	0	1	0	1
	11	1	0	1	0
	10	0	1	0	1

$(A=0)$

	DE	00	01	11	10
BC	00	0	1	0	1
	01	1	0	1	0
	11	0	1	0	1
	10	1	0	1	0

(a) $A \oplus [B \oplus C \oplus D \oplus E]$ (b) $A \oplus B \oplus C \oplus D \oplus E$ (c) ~~Both~~ (d) None

★

$$A [B \oplus C \oplus D \oplus E] + \bar{A} [B \oplus C \oplus D \oplus E]$$

$$= \bar{A}B + A\bar{B} = A \oplus B$$

$$= A \oplus B \oplus C \oplus D \oplus E // [C].$$

★ Relations :-

$$\Rightarrow \overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B}$$

$$\Rightarrow A \oplus B \oplus C = A \oplus B \oplus C$$

$$\Rightarrow \overline{A \oplus B \oplus C} = A \oplus B \oplus C = A \oplus B \oplus C = C \oplus A \oplus B$$

$$\Rightarrow \overline{A \oplus B \oplus C \oplus D} = \bar{A} \oplus (\bar{B} \oplus \bar{C} \oplus \bar{D}) = A \oplus (B \oplus C \oplus D)$$

$$= \overline{A \oplus B} \oplus [C \oplus D] = A \oplus B \oplus C \oplus D$$

$$= A \oplus B \oplus C \oplus D //$$

$$= A \oplus B \oplus C \oplus D //$$

$$\begin{aligned} \Rightarrow A \oplus B \oplus C \oplus D \oplus E &= A \oplus B \oplus C \oplus D \oplus E \\ \Rightarrow \overline{A \oplus B \oplus C \oplus D \oplus E} &= \overline{A \oplus B \oplus C \oplus D \oplus E} \\ &= A \oplus B \oplus C \oplus D \oplus E \\ &= A \oplus B \oplus C \oplus D \oplus E \\ &= A \oplus B \oplus C \oplus D \oplus E \end{aligned}$$

(a) $Y = A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G$ will be equivalent to

(a) $A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G$

(b) $A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G$

(c) $A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G$

(d) $A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G$

A. $Y = \overline{A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G} = \overline{A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G}$

(b) $\overline{A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G} = \overline{A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G}$

(c) $A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G = \overline{A \oplus B \oplus C \oplus D \oplus E \oplus F \oplus G}$

(*) Implicants, Prime Implicants, Essential Prime Implicants, Redundant Prime & Selective Prime Implicants :-

Boolean Expression

Implicant

Prime Implicant

(*) ~~EPI~~ ~~RPI~~ ~~SPI~~

Implicants :- Subsets of B. function.

Ex:- $f(A, B, C, D) = \sum m(0, 1, 4, 6, 9, 12, 15)$

$\rightarrow g_1 = \sum m(1, 9, 12, 15) (\checkmark)$

$\rightarrow g_2 = \sum m(0, 6, 12) (\checkmark)$

$\rightarrow g_3 = \sum m(0, 1, 3, 4) (X)$

$f(A, B, C, D) = \sum m(0, 1, 5, 7, 10, 11, 14, 15)$

(✓) $\rightarrow g_1 = AC = ABCD + \overline{A}BCD + ABC\overline{D} + \overline{A}B\overline{C}\overline{D} = \sum m(10, 11, 14, 15)$

(✓) $\rightarrow g_2 = \overline{A}BD + \overline{A}\overline{B}\overline{C} = \overline{A}BCD + \overline{A}B\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D = \sum m(7, 10, 5)$

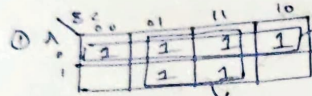
(X) $\rightarrow g_3 = (A + B + \overline{C}) = \sum m(2, 3)$

$= (A + B + \overline{C} + 0)(A + B + \overline{C} + 0) = \sum m(0, 1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$

→ Prime Implicant: Best Possible Group.

Process: ① Start with Best " " " " [8 cells, 4 cells, 2 cells...]
 ② Inside the bigger club, no further small clubs allowed.

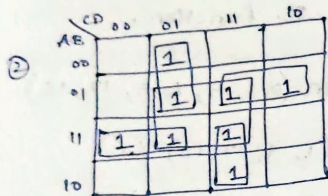
$$f = \sum m(0, 1, 2, 3, 5, 7)$$



Quad 1

⇒ 2 quads - 2 P.I

$$\Rightarrow \sum m(1, 3, 5, 7)$$



⇒ 4 pair + 1 quad = 5 Prime

Implicants

$$\Rightarrow \sum m(5, 7, 13, 15)$$

$$\Rightarrow \sum m(1, 5), \sum m(6, 7),$$

$$\sum m(12, 13), \sum m(14, 15)$$

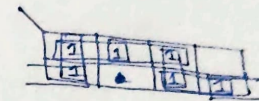
Max. Prime Implicants:

2-Variable:-



→ 2 P.I

3-Variables:-



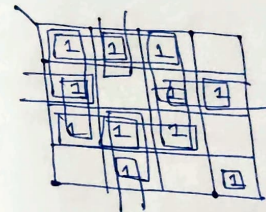
→ 6 P.I

$$\sum m(0, 1), \sum m(1, 2), \sum m(0, 4)$$

$$\sum m(6, 7), \sum m(4, 5),$$

$$\sum m(3, 7)$$

4-Variables:-



→ 13 P.I

END OF
DAY-11